Deployment Environment and Cloud Platform Recommendation

Conveyor Predictive Maintenance

INFT 41000
Deploying Artificial Intelligence Systems
Assignment 1

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This report uses a realistic plant scenario. Assumptions are identified in the body; no plant dataset, model test, or deployment has been independently verified.

GitHub repository and commit

https://github.com/srijan2419/INFT41000-Deploying-Artificial-Intelligence-Systems-Assignment-1

View committed version

AI-use statement

OpenAI ChatGPT/Codex was used within the permitted scope to explain concepts, research official documentation and prices, assist with the analysis, scorecard ratings and cost estimates, and format this report. I supplied the plant scenario and the preference for hybrid deployment and Azure, and reviewed the report. Service names and prices were checked against provider documentation and Microsoft's retail pricing API. Calculations and assignment requirements were checked, and scenario assumptions are identified in the report.

1 Use case and readiness snapshot

1.1 Use case

The use case is predictive maintenance for five conveyors at a manufacturing plant in southern Ontario. The conveyors run 24 hours a day, seven days a week. Their sensor readings are collected every second, and three months of history and reasonably accurate breakdown and repair records already exist.

The proposed inputs are motor or bearing temperature, vibration summaries, and motor current. These are common condition-monitoring signals [5]. The vibration input is a summary produced by a sensor, not a raw vibration waveform sampled only once a second. A small classification model would estimate whether a conveyor is at increased risk of a fault in the next 24 hours. Security and the manager receive alerts; this proposal adds a maintenance technician who can inspect the equipment. Monitoring and feedback matter most, because equipment wear and repairs can change the relationship between readings and faults.

1.2 Production requirement

A notebook prototype cannot provide alerts throughout a plant's continuous operating schedule. Missed faults may cause unplanned downtime, while repeated false alerts can waste inspection time, so the service needs monitoring, recovery, and a clear response process [1].

1.3 Readiness summary

The six-dimension assignment template is used [3], rather than the separate seven-dimension notebook exercise. Appendix A contains the completed scorecard and the deployment effects.

Dimension	Score / 5	Gap
Data	2	Fix before launch
Model	0	Fix before launch
Infrastructure	1	Fix before launch
Organizational	2	Fix before launch
Risk	3	Fix before launch
Cost	3	Fix before launch

Total: 11/30. **Average:** 1.83/5. These are draft judgments from the stated scenario and gaps, not audit results.

1.4 Two weakest dimensions

Model readiness is 0 because no model or test results exist. Infrastructure is 1 because collecting readings does not prove that scoring, monitoring, and recovery work. These gaps favour a limited pilot using managed training and a tested local scorer; they do not support immediate production launch.

2 Deployment environment evaluation

2.1 Environment Fit Scorecard

The workload scored is the demanding part: live conveyor alerting. The cloud training workload is considered separately. Table B1 and the completed scorecard in Appendix B contain the detailed evidence [4].

Question in order	Score / 5	Main reason
Data sensitivity	2	Raw plant and repair data stay local under the assumed policy.
Cost pattern	3	Live scoring is steady; retraining is occasional.
Operational capacity	4	Existing plant IT has no dedicated ML operations team.
Latency	1	Live alerts must continue locally during a WAN outage.

Average: (2 + 3 + 4 + 1) / 4 = 2.5. **Spread:** 4 - 1 = 3.

First, the latency score of 1 rules out public cloud alone for live alerting. Second, the spread of 3 points toward hybrid. The average is in the mixed range, but it does not override those earlier rules. The scorecard therefore agrees with hybrid [4].

2.2 Comparison

On-premise provides local alerting and control, but the plant must also run its training tools. Cloud can provide managed training and capacity, but a cloud-only scorer depends on the internet connection. Hybrid uses local scoring and rented batch training, with extra work to maintain the connection between them. This separates streaming-style alerts from batch training, following the Week04 pipeline comparison [18]. Table B2 compares all five required dimensions for this system.

2.3 Recommendation

Recommendation: use hybrid deployment, keeping live scoring and sensitive raw records on-premise while using Azure for approved training data, model retraining, and model management.

Latency is the deciding question: the alert path must still work when the WAN fails. Data sensitivity also favours keeping raw readings and identifying repair records at the plant. Local processing handles the small, continuous scoring workload. Azure handles occasional CPU training and longer-term comparisons, where waiting is acceptable. Approved model files are downloaded and checked before replacing the local version. Operational capacity supports managed cloud training because there is no dedicated ML operations team. Frontier language models are not needed to classify conveyor sensor patterns. The scorecard agrees with this recommendation.

2 Deployment environment evaluation continued

2.4 Three-year cost comparison

This is an incremental operating estimate in USD, excluding tax and price changes. USD avoids mixing Microsoft's published currency with an assumed exchange rate. Canada Central Linux pay-as-you-go prices were retrieved on October 9, 2026 [12]. No reservation, Spot, or student credit is assumed.

Five conveyors produce five combined rows per second: 432,000 rows per day and 38.88 million over 90 days. At an assumed 100 bytes per row, the history is about 3.9 GB before file overhead. Local data is retained for 90 days. Cloud storage budgets are 20 GB for hybrid features and models, and 50 GB for the cloud alternative's raw data and models. A daily export includes one-minute summaries and fault labels.

Both options use eight billed hours per month of Standard_DS3_v2 training, including startup and idle overhead. Hybrid reuses the existing sensor PC, subject to a capacity check. The cloud-only alternative adds one Standard_D2s_v3 scoring VM, running 26,280 hours over three years, and a 32 GiB E4 disk. These are basic single-host estimates, not the cost of a redundant high-availability design.

Three-year cost line	Hybrid USD	Cloud-only USD
Compute	80.64	2,997.72
Storage, disk and ancillary allowance	374.40	491.04
Registry and Key Vault	183.51	183.51
Setup and operating staff	16,560.00	12,420.00
Allocated local scoring power	157.68	0.00
Total	17,356.23	16,092.27

Staff assumptions are USD 45/hour: 80 setup hours and eight operating hours/month for hybrid, versus 60 and six for cloud. Existing sensor acquisition, the gateway, and normal maintenance duties are common baseline costs and are excluded. Hardware reuse, electricity, labour, and the USD 10/month ancillary allowance are planning assumptions, not vendor quotes. Appendix C shows the unit rates and arithmetic.

Cloud-only is about USD 1,264 cheaper, but it fails the offline alert requirement. Staff time is the most uncertain line: hybrid operation at four to sixteen hours/month changes its total to roughly USD 10,876 to USD 30,316. Incident support or a new local PC would increase the estimate. Avoided downtime has not been measured, so this is not a proven return on investment.

2.5 Tipping point

If approved features and fault labels are no longer allowed outside the plant network, I would switch to fully on-premise training and scoring. That is a specific data-control limit, rather than a general preference against cloud.

3 Cloud platform and service selection

3.1 Why Azure

Azure is selected for three reasons. Managed training and model tracking reduce the tools the small team must maintain [6]. Blob Storage, Functions, and monitoring fit the export and training workflow on one platform [8, 9, 14]. Training clusters can scale to zero nodes between short jobs [7]. Canada Central is the proposed region; service availability, quotas, and the scope of Canadian data residency still need checking before building.

Why not AWS: although the notebooks use AWS, this proposal favours Azure's integrated managed-training workflow. **Why not Google Cloud:** its ML tools could work, but this case has no specific need that outweighs the chosen Azure workflow.

3.2 Service map

Table C1 gives the exact service names. Machine Learning handles training; Blob Storage holds approved files; Functions provides status and job submission. Container Registry, Key Vault, and Azure Monitor support images, secrets, and cloud monitoring [6-11, 14]. The live Docker scorer stays local.

The managed services are PaaS: the plant configures applications and data, while Microsoft runs the service infrastructure. Directly operating the alternative Azure VM is IaaS. The local PC is outside these cloud models [2].

3.3 Containers and serverless

Component	Traffic, duration, then portability	Choice
Live scorer	Steady five rows/second; continuous process; must run locally with the same dependencies.	Container
Cloud training	Occasional jobs; may take hours; use a versioned Python environment.	Container job on managed compute
Status API / job submission	Low, uneven request volume; short calls; can remain Azure-specific.	Functions Flex Consumption

Containers package the software; serverless describes compute management. The scorer needs a persistent local process, while short cloud calls can tolerate a cold start. Training runs outside the Function after submission [7, 9].

3.4 Build or buy

Managed AutoML can try model candidates quickly, but still needs clean labels and independent testing [19]. A self-managed Python stack gives full control but requires maintaining training tools, job tracking, and a registry. Managed services add usage charges; self-management adds staff costs. Both require data and evaluation expertise. I would build a small plant-specific classifier on Azure Machine Learning managed training, buying platform support rather than building an ML operations platform. AutoML can provide a pilot comparison; its result still needs testing.

3 Cloud platform and service selection continued

3.5 Cloud Adoption Framework

Plan: agree on the fault definition, response, data policy, and budget. Measure existing downtime to compare the value of alerts with their cost.

Ready: create separate development and production resource groups, restricted identities, and Canada Central storage with anonymous access disabled. Set budget alerts and export/model approval owners before building.

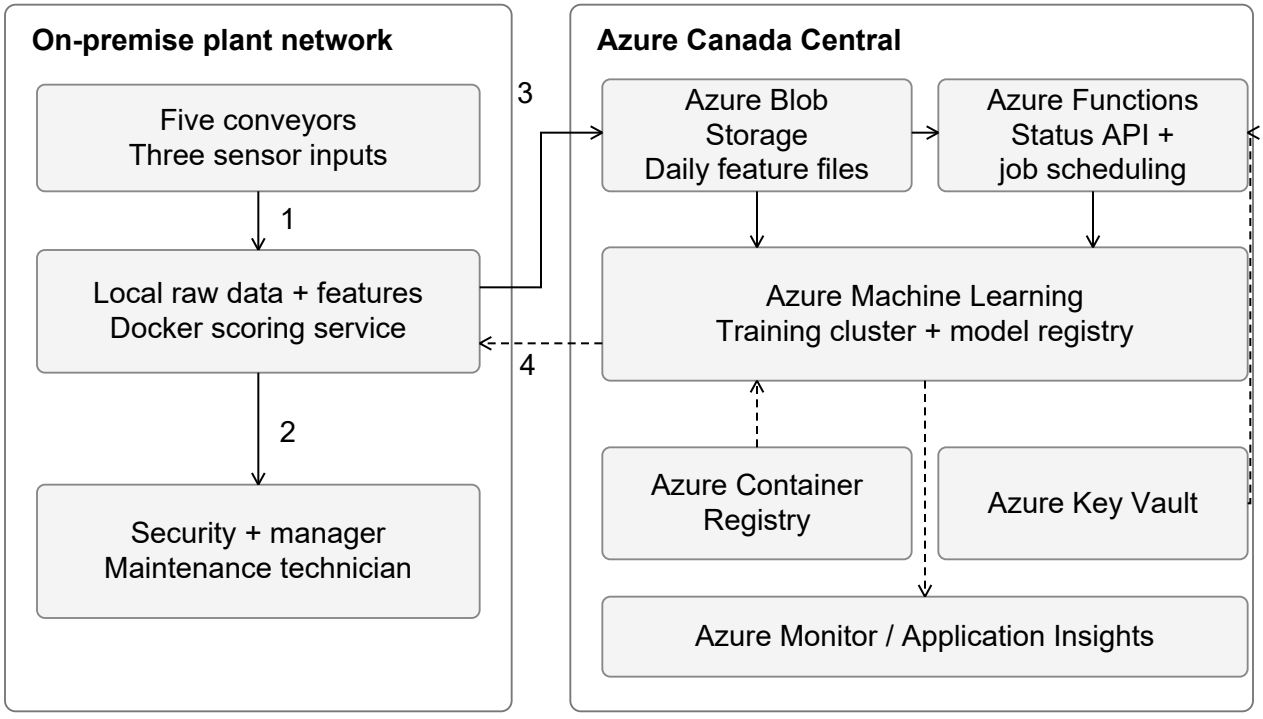
Adopt: pilot one conveyor in shadow mode: recommendations do not change operation. Test exports, training, and rollback before adding the other four.

Govern & Operate: review access and spending monthly; patch the PC; check missed and false alerts; retain the previous model. Name maintenance and IT owners and test recovery. These apply the assignment's phases using the Azure framework [15].

3.6 Lock-in

Lock-in is strongest in Azure jobs and registry metadata, Function bindings, and permissions. Keep training code and dependency versions portable, export features as CSV or Parquet, and test the model in the local container. Migration would still require replacing cloud permissions and orchestration.

Figure 1 Deployment diagram



Legend: solid arrows show readings, alerts, exports, and jobs; dashed arrows show approved models or supporting services. 1: one-second readings. 2: local alerts. 3: daily approved features and labels over HTTPS. 4: approved model download. Raw records and the local alert path remain inside the plant.

4 Risks and decision

4.1 Three main risks

Risk	Likelihood / impact	Specific mitigation
Missed faults or unavailable local scoring	High / High	Test independent faults and a 30-day shadow pilot. Keep existing safety controls. Warn when input or scoring stops; test recovery on a spare PC.
Data exposure across the hybrid boundary	Medium / High	Remove identifying repair details before export. Restrict access, disable anonymous storage, use HTTPS, and review permissions. Keep cloud credentials separate from the scorer.
Growing cloud and support costs	Medium / Medium	Set minimum training nodes to zero, cap runtime, and review monthly spending and support hours. Reassess services if demand becomes continuously large.

The Week03 notebook uses Dropbox's Magic Pocket as a real example of scale changing the environment decision [16]. Dropbox describes building its own storage infrastructure [17]. This supports reassessing costs when workloads change.

Under shared responsibility, Azure runs PaaS infrastructure; the plant owns its model, permissions, export rules, and response. Local PC patches and recovery also remain its responsibility [2, 16].

4.2 Decision

Decision: Delay production launch; proceed only with a supervised feasibility pilot. The 11/30 readiness score includes no model and no tested scoring deployment.

Go requires validated data and fault labels; evaluation on a later period and a held-out conveyor with at least 20 independent fault events; a 30-day shadow pilot with proposed recall of at least 90% and no more than one false alert per conveyor per week; alerts within two seconds during a WAN outage with tested recovery and rollback; and named maintenance and IT owners with approved permissions and budget. These are proposed targets, not achieved results. More history is needed if too few labelled faults exist.

Security forwards alerts, the manager coordinates priorities, and maintenance inspects the conveyor. Existing safety controls remain independent. The manager checks the evidence before changing Delay to Ship.

5 Communication and professional practice

Figure 1 matches Table C1. Appendices A-C contain the scorecards and supporting tables. Numbered citations refer to the following references. The cover records version evidence and the actual AI assistance.

5 Communication and professional practice

References

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5 Communication and professional practice continued

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Appendix A Completed Readiness Scorecard

Production Readiness Scorecard - six dimensions, scored 0 to 5

Use case: Conveyor predictive maintenance
Prepared by: Srijan Choudhary **Date:** October 9, 2026

Scale: 0 = nothing exists/cannot assess; 1 = red flag; 2 = between 1 and 3; 3 = developing; 4 = between 3 and 5; 5 = ready. Scenario assessment; assumptions are not verified plant evidence.

Dimension	Rubric anchors	Score	Evidence	Gap
Data	1 No validation, unknown lineage 3 Validated; lineage undocumented 5 Validated, versioned, governed	2/5	Three months of one-second readings and repair records are available in the scenario, but missing-data checks and versioned labels have not been demonstrated.	Fix before launch
Model	1 Never tested outside training data 3 Tested once; no bias/drift plan 5 Tested, explainable, drift monitored	0/5	No trained conveyor model or independent test results have been supplied.	Fix before launch
Infrastructure	1 Runs on someone's laptop 3 Deployed; no monitoring/rollback 5 Containerized, monitored, autoscaled, recoverable	1/5	Sensor collection exists, but a monitored scoring service, tested recovery, and rollback do not yet exist.	Fix before launch
Organizational	1 No named owner 3 Owner; no documented process 5 Owner, on-call, governance documented	2/5	Security and the manager receive alerts, but maintenance escalation and ML support ownership are not documented.	Fix before launch
Risk	1 No risk review 3 Risks identified; not mitigated 5 Identified, mitigated, monitored	3/5	This report identifies safety, access, and cost risks, but their controls have not been implemented or formally reviewed.	Fix before launch
Cost	1 Cost never estimated 3 Estimated; not compared to value 5 Tracked and justified by value	3/5	A three-year operating estimate is included, but avoided downtime and the value of the alerts have not been verified.	Fix before launch

Total: 11/30 **Average:** 1.83/5
Lowest: Model 0, Infrastructure 1 **Any 0 or 1:** Yes

Decision: DELAY for remediation. Ship and Kill are not selected.

Conditions and next actions: Maintenance lead and IT owner to validate data, test the model, prove offline alerting and recovery, and approve access and budget before launch. Target review: 23 October 2026; the 30-day shadow pilot follows successful prechecks. These are proposed roles and dates.

Reviewed by: Pending review by the plant manager and maintenance lead. No sign-off has occurred.

This completed one-page scorecard is excluded from the four-page appendix allowance [3].

Appendix A Readiness and deployment effects

This table supplements the completed Readiness Scorecard. All gaps require work before full production launch. Scenario facts are student-supplied; statements about untested components describe the absence of evidence, not a site audit.

Dimension	Score	Evidence	Effect on deployment
Data	2	Three months of one-second readings and repair records are available in the scenario, but missing-data checks and versioned labels have not been demonstrated.	Keep the original readings locally and validate the daily training export before Azure uses it.
Model	0	No trained conveyor model or independent test results have been supplied.	Use managed training for a pilot, with local scoring only after the model passes its tests.
Infrastructure	1	Sensor collection exists, but a monitored scoring service, tested recovery, and rollback do not yet exist.	Test local alerting during a WAN outage and prove that the current PC can handle scoring.
Organizational	2	Security and the manager receive alerts, but maintenance escalation and ML support ownership are not documented.	Name a maintenance lead and an IT support owner instead of assuming Azure removes these responsibilities.
Risk	3	This report identifies safety, access, and cost risks, but their controls have not been implemented or formally reviewed.	Keep AI advisory and separate from the existing safety controls.
Cost	3	A three-year operating estimate is included, but avoided downtime and the value of the alerts have not been verified.	Approve the pilot budget and measure downtime before committing to a full rollout.

Facts and assumptions

Student-supplied facts: southern Ontario; five conveyors; continuous operation; readings every second; three months of history; reasonably accurate repair records; alerts to security and the manager; a preference for hybrid and Azure.

Planning assumptions: temperature, vibration summaries, and current are available; an existing sensor PC can be reused after a load test; approved summaries may leave the network; there is no dedicated ML operations team; maintenance staff join the response process. The two-second alert target, retention policy, support hours, and performance gates are proposed requirements.

No actual data quality report, trained model, model accuracy, current hardware inventory, staffing roster, formal risk sign-off, or downtime-value calculation has been supplied. These must be checked before launch.

Appendix B Completed Environment Fit Scorecard

Four questions scored 1 to 5 - on-premise, cloud, or hybrid

Use case: Live conveyor alerting
Prepared by: Srijan Choudhary **Date:** October 9, 2026

Scale: 1 = strong on-premise pull; 2 = leans on-premise; 3 = mixed; 4 = leans cloud; 5 = strong cloud pull. Scores apply to the stated scenario, not verified plant measurements.

Question in order	Score	Evidence
1 Data sensitivity	2/5	Assumed plant policy keeps raw operational readings and identifying repair records on-site, while approved summaries may leave the network.
2 Cost pattern	3/5	Five conveyors create a steady live workload, but model retraining is an occasional batch job.
3 Operational capacity	4/5	Assumed plant staff can maintain the sensor PC, but have no dedicated ML operations team to manage a cloud training stack.
4 Latency	1/5	The proposed live-alert path must run on-site and continue during a WAN outage; its target is within two seconds of receiving a reading.

Average: 2.5/5 **Spread:** 3
Data or latency scored 1: Yes - latency.
Ruled out: Public cloud alone for live scoring.

Decision rule in the required order

A Hard constraint: data sensitivity or latency = 1 rules out public cloud alone for that workload. This applies here.
B Wide spread: a spread of 3 or more means hybrid with a workload split. This also applies.
C Otherwise average: 2.4 or less means on-premise; 3.6 or more means cloud; between means either, with operational capacity breaking the tie. This step does not override A or B [4].

Recommendation: HYBRID. On-premise and Cloud alone are not selected.

One-sentence justification: Keep live alerting and raw data at the plant so the system works offline, and use Azure managed services for occasional retraining on approved features.

Question that drove the decision: Latency; operational capacity supports managed cloud training.

This completed one-page scorecard is excluded from the four-page appendix allowance [4].

Appendix B Environment summary and comparison

Table B1 Environment Fit Scorecard summary

Question in order	Facts about this use case	Score
1 Data sensitivity	Assumed plant policy keeps raw operational readings and identifying repair records on-site, while approved summaries may leave the network.	2
2 Cost pattern	Five conveyors create a steady live workload, but model retraining is an occasional batch job.	3
3 Operational capacity	Assumed plant staff can maintain the sensor PC, but have no dedicated ML operations team to manage a cloud training stack.	4
4 Latency	The proposed live-alert path must run on-site and continue during a WAN outage; its target is within two seconds of receiving a reading.	1

Average: 2.5. **Spread:** 3. **Data/latency = 1:** Yes.
Rule: Hybrid. **Recommendation:** Hybrid.

Table B2 Environment comparison

Dimension	On-premise	Cloud	Hybrid
Data control	All readings, repair details, and models stay on the plant network.	Raw readings and repair data would need approval to leave the plant; Azure permissions still need local management.	Raw readings stay local; only approved features, fault labels, and model files enter Azure.
Cost model	The plant owns training and serving capacity even when retraining is idle.	Training is rented, but live scoring needs continuously available cloud capacity.	Reuse the local PC for scoring and rent CPU training for short jobs.
Scalability	Adding conveyors or larger training jobs depends on spare local capacity.	Cloud training and serving capacity can increase, subject to quotas and budget.	Training can grow in Azure; local scoring still needs capacity testing as conveyors are added.
Operational burden	Plant staff handle operating-system updates, training tools, monitoring, and recovery.	Azure manages PaaS infrastructure; the plant still owns its model, data access, and alert response.	Staff maintain one local scorer plus the export and model-update process; Azure manages training infrastructure.
Best fit for this system	Supports offline alerts, but requires more in-house training support.	Simpler central hosting, but loses timely scoring if the WAN fails.	Fits the offline requirement and the small team, with added integration work.

Appendix C Service map

Table C1 Service map

Cloud platform: Microsoft Azure. The on-premise scorer is included to make the hybrid split explicit; it is not labelled as an Azure service.

Building block	Named service	Model	Container / serverless	Why
Training compute	Azure Machine Learning compute cluster; Standard_DS3_v2	PaaS	Container job	Managed CPU training; minimum zero nodes between jobs.
Live serving / hosting	On-site Python model in Docker on the existing sensor PC	Not a cloud service	Container	Scores each conveyor locally and works without WAN access.
Cloud serving / hosting	Azure Functions, Flex Consumption	PaaS	Serverless	Provides a small status API and submits scheduled training jobs.
Data storage	Azure Blob Storage, Standard Hot LRS	PaaS	Not applicable	Stores approved training files and exported model files.
Managed AI / ML	Azure Machine Learning workspace and model registry	PaaS	Managed platform	Tracks training runs, model versions, and approval records.
Container images	Azure Container Registry, Basic	PaaS	Image registry	Stores versioned images; the local scorer keeps a cached approved image.
Secrets	Azure Key Vault, Standard	PaaS	Not applicable	Stores cloud-side certificates and secrets with restricted access.
Monitoring	Azure Monitor with Application Insights	PaaS	Not applicable	Records cloud failures and job status; local alarms remain local.

Azure service names and roles are supported by [6-11, 14]. The managed training cluster is treated as PaaS even though it uses VM-sized nodes. The alternative directly operated Azure VM is IaaS; no SaaS product is required by this design [2].

The local collector makes features and uploads a daily file over HTTPS. A timer Function submits an Azure Machine Learning training job; it does not run the long training job itself. A separately authenticated HTTP Function can return the stored status summary. Approved models and a compatible image are downloaded before deployment; the scorer uses cached local copies if the WAN is down.

Appendix C Three-year cost detail

Table C2 Three-year cost comparison

Option A: Hybrid. **Option B:** Cloud-only scoring and training.
Currency: USD. **Region:** Canada Central. **Access date:** October 9, 2026.

Cost line	Assumption and source	A USD	B USD	Uncertainty
Compute	Training: 288 hours x $0.28 = $80.64 in both. B adds 26,280 hours x $0.111 = $2,917.08 for Linux D2s_v3. A reuses a local PC. Retail API [12].	80.64	2,997.72	Medium
Storage and data transfer	Hot LRS: A 20 GB, B 50 GB x $0.02 x 36 months. B E4 disk: $2.64 x 36 = $95.04. Both add $10 x 36 for training disk, logs, operations and small network charges. Rates [12]; allowance is an assumption.	374.40	491.04	Medium
Managed services / licences	ACR Basic: $0.1666/day x 1,095 = $182.427. Key Vault: 10,000 operations/month x $0.03 x 36 = $1.08. ML adds no separate platform fee [13]. Functions assumed below Flex free grants [9]. Rates [12].	183.51	183.51	Low
Staff time to operate	Assumed $45/hour. A: 80 setup + 8 x 36 operating hours = 368 hours. B: 60 + 6 x 36 = 276 hours. Same baseline plant duties excluded; no vendor labour quote.	16,560.00	12,420.00	High
Local scoring power	A allocates 0.05 kW x 26,280 hours x assumed $0.12/kWh. B has no added local scorer. Electricity rate is a budget assumption, not an Ontario utility quote.	157.68	0.00	Medium
Three-year total	Sum of the five lines. No tax, inflation, credits, or new sensor purchase. No redundant-host design is priced.	17,356.23	16,092.27	High

Scope and uncertainty

Hardware acquisition is $0 only under the explicit reuse assumption. New hardware, a redundant scoring host, premium networking, higher log retention, or round-the-clock external incident cover would need a revised budget. The ancillary allowance is a reserve, not a provider list price. No continuous VPN or private endpoint is assumed; exports use authenticated HTTPS and restricted storage access.

For Functions, assume fewer than 2,000 short invocations and 6,000 GB-seconds per month, no always-ready instances, and an eligible paid subscription whose free grant is available. The function storage account and other small charges are covered by the ancillary allowance; they are not claimed to be free.

Retail API evidence is saved beside this report as price-evidence.json. Staff-time sensitivity in Section 2.4 changes operating hours only; setup hours stay fixed. Savings from fewer breakdowns remain unverified.